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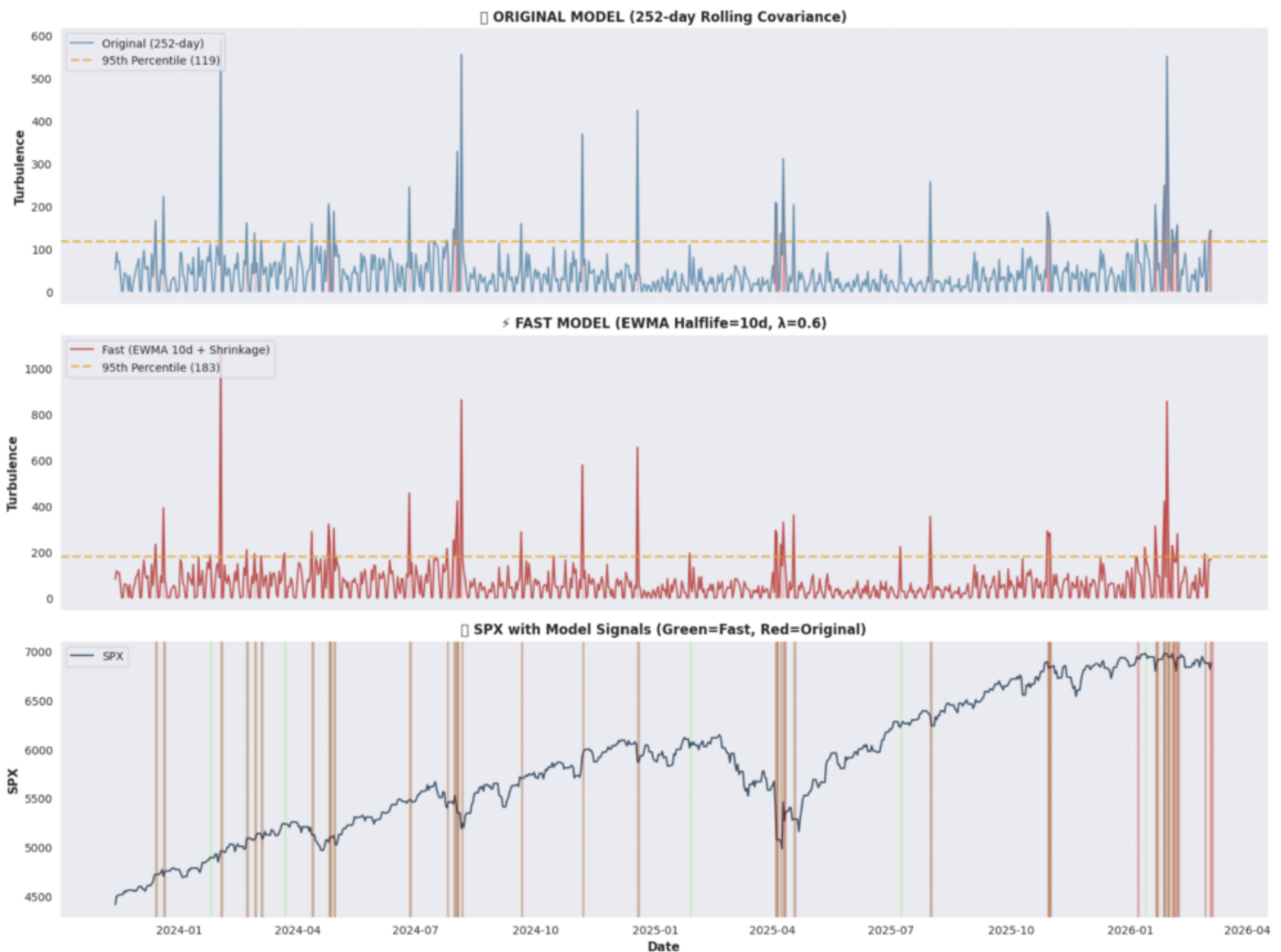
Round 1, Fight 1

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Turbulence Signals in the AI Disruption Regime



Source: 22V Research

I have received a number of inquiries about the turbulence model, so I thought the easiest way to explain what is happening was to put together a short note. I also added a couple of new features to the framework to better reflect the environment I believe we are entering, which I discussed in the paper I sent out this morning on the structural changes emerging from the AI disruption regime.

The top chart shows the original version of the model. You can see the large spike in early February, which occurred after three consecutive days of elevated turbulence readings at a time when the S&P 500 was still above its 50-day moving

average and the VIX remained low. The model uses a 252-day (one-year) rolling covariance approach, which measures how unusual current cross-asset movements are relative to the past year. It is difficult to see on the chart, but the last three days have also produced three consecutive elevated readings. For those watching their PnL recently, that likely does not come as a surprise.

Because I expect the coming environment to feature higher volatility of volatility, I also introduced a second version of the model that uses an exponentially weighted covariance approach. This places more emphasis on recent market behavior, allowing the model to adapt more quickly to changes in correlations and volatility. The system therefore now measures market stress in two different ways simultaneously. The original model examines how assets across the global market normally behave relative to each other over roughly the past year and then asks whether today's movements are unusual compared with that longer history. Because it uses a full year of data, it reacts slowly and produces relatively few signals, but those signals tend to represent meaningful structural stress rather than short-term noise. In the chart, this appears as occasional large spikes that cross the threshold line. When that happens, it means relationships across equities, bonds, commodities, currencies, and volatility have shifted in a way that has been rare over the past year.

The new version of the model operates much faster because it emphasizes what has happened in the most recent days and weeks rather than treating the past year equally. This makes it more sensitive to sudden changes in market behavior. As a result, the turbulence readings in the second panel spike more frequently and more dramatically because the model reacts quickly to shifts in volatility, sector rotations, and correlation changes across markets. These signals often appear before the slower model detects anything, which means the fast model can highlight early instability that is just beginning to develop. At the same time, because it is more sensitive, it also produces temporary signals that do not always lead to major market moves.

When these signals are overlaid on the S&P 500 in the bottom panel, a clear pattern emerges. The fast model typically flashes warnings first, while the slower model confirms them later if the instability persists. In that sense, the fast model acts as an early detection system, while the original model serves as confirmation that the disturbance has become more significant. Recently, the chart shows a cluster of signals from the fast model and several signals from the slower model as well, even though the S&P 500 remains near its highs. That combination suggests the internal relationships between markets are becoming less stable beneath the surface, which is precisely the type of environment this turbulence framework is designed to identify.

Two things stand out to me when thinking about the current regime. First, we have seen more elevated turbulence readings than at any point over the past three years. Second, historically when clusters of elevated readings occurred, the S&P 500 had usually already declined. Today that is not the case as we still sit near the all-time highs. This dynamic aligns with the environment I have been writing about this year. It is why I titled this note **Round 1, Fight 1**. Once Rocky Balboa knocked down Apollo Creed, we knew it was going to be a long fight. Investors should prepare for the same kind of environment ahead.

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